

**CREATOR
PULSE**

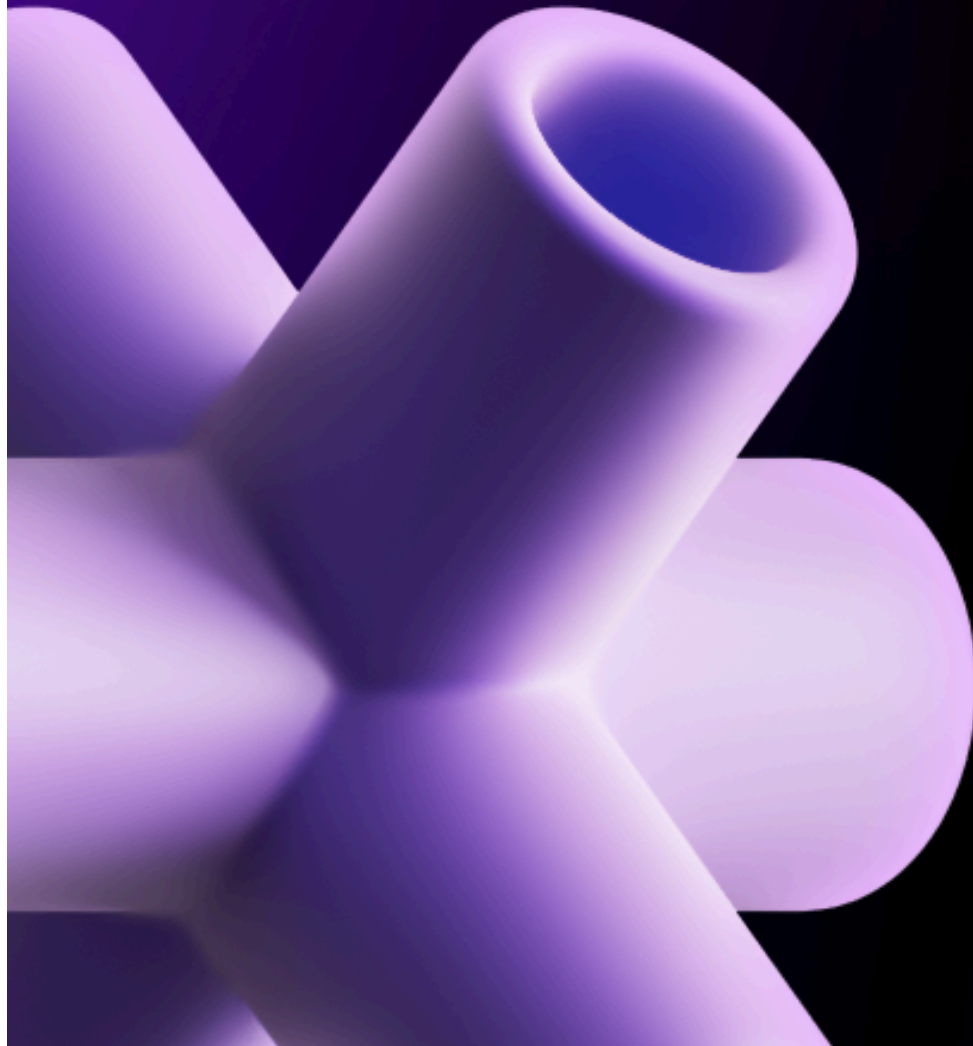


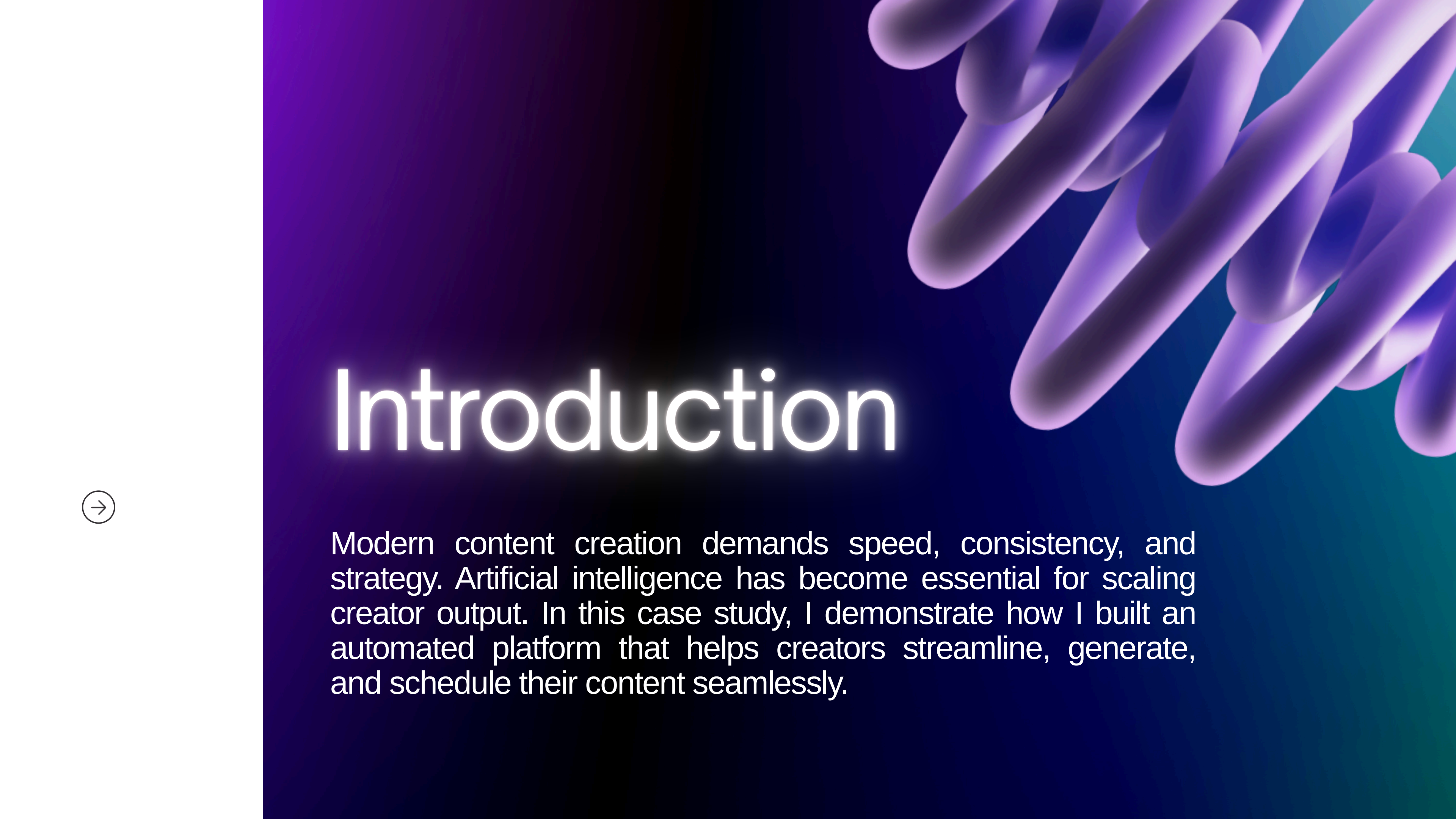
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**02 Oct,
2026**

App Case Study

CREATION MADE EASY



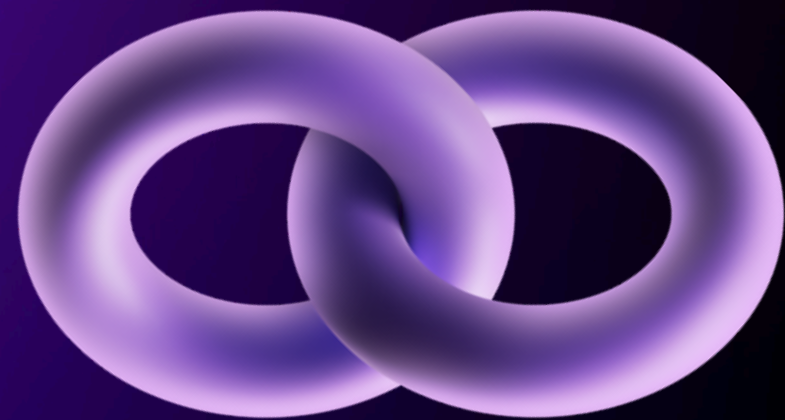


Introduction



Modern content creation demands speed, consistency, and strategy. Artificial intelligence has become essential for scaling creator output. In this case study, I demonstrate how I built an automated platform that helps creators streamline, generate, and schedule their content seamlessly.

Project Overview



Name

CreatorPulse

Next-Gen Creator Workflow & AI Studio

**Target
Users**

Content Creators,
Influencers & Technical
Writers

Field

Creator
Economy / AI
SaaS

Goal

Bridge performance analytics
with AI generation and smart
scheduling to turn creator
data into automated content
momentum

Issues Faced

Fragmented Creator Tooling

Creators waste time juggling separate apps for analytics, script writing, email dispatch, and scheduling.

Generic AI Content Outputs

Standard AI tools generate bland text that ignores platform constraints, video hooks, and target durations.

Guesswork in Posting Schedules

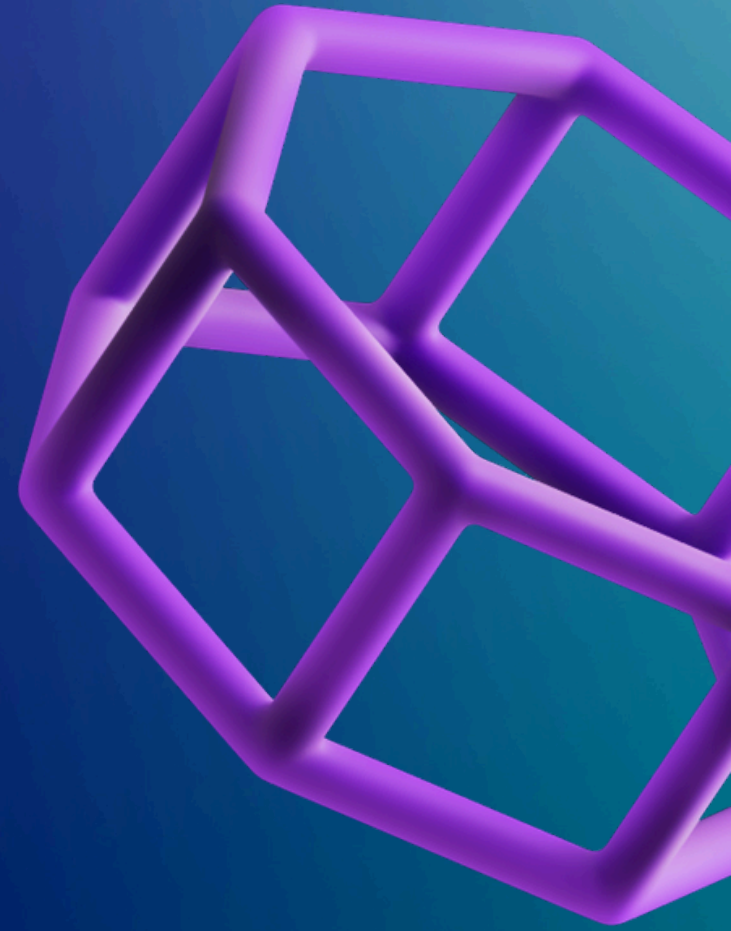
Creators post blindly without knowing when their audience is active or what video lengths retain viewers.

Disconnected Audience Outreach

Managing social posts and newsletter campaigns separately leads to inconsistent brand messaging.



AI-Implemented Solutions



Multimodal Content Engine

Generates platform-tailored short video scripts, SEO blogs, newsletters, and captions with customizable tone profiles.

Predictive Analytics Parser

Analyzes TikTok, IG, and YouTube CSV exports to identify optimal post times, ideal video durations, and peak engagement slots.

Smart Calendar & Golden Windows

Overlays peak audience activity heatmaps onto a drag-and-drop schedule to automate posting windows.

Built-In Email Dispatch & Library

Composes and sends newsletters directly to subscriber lists while organizing all drafts in a single workspace.

Key Prompt Engineering Techniques Used

Technique	Implementation in CreatorPulse
System Persona Prompting	Adopting specialized roles (Viral Scriptwriter, Technical Copywriter, Email Marketer) to set behavioral boundaries.
Dynamic Few-Shot Injection	Injecting top-performing creator hooks and formatting examples into context based on the selected platform.
Negative Prompting / Constraints	Explicitly listing forbidden phrases ("In today's fast-paced world...", "Hey guys", "Unleash your potential").
Structured Output Schema	Enforcing strict Markdown schemas to enable seamless rendering in the CreatorPulse UI components
Word-to-Duration Mapping	Programmatically calculating target word limits based on speech pacing mathematics ($\text{\text{Words}} \approx \text{\text{Seconds}} \times 2.5$).

Core Features & Platform Capabilities

Feature 1: Multimodal AI Generator

Short-Form Video Scripts:

Complete with 3-second visual hooks, audio timestamps, and duration targets.

SEO Blogs & Newsletters:

In-depth, structured long-form writing optimized for readability and audience conversion.

Social Captions & Code Snippets:

Platform-native formatting, hashtag strategy, and step-by-step code walkthroughs.

Feature 2: Predictive Analytics Engine

Data Parser:

Accepts CSV and JSON exports directly from TikTok, Instagram, and YouTube Studio.

Metrics Extraction:

Identifies optimal posting hours, peak engagement days, top-performing formats, and ideal video lengths.

Golden Window Heatmaps:

Highlights exact time slots with maximum audience activity.

Feature 3: Smart Content Calendar & Workflow

Status Tracking:

Tracks content progression through Idea $\rightarrow$ Scripted $\rightarrow$ Scheduled $\rightarrow$ Published.

Drag-and-Drop Scheduling:

Easily align drafts with your analytics-driven Golden Windows.

Platform Tagging:

Color-coded organization across YouTube, TikTok, Instagram, and Substack.

Feature 4: Built-In Email Dispatch & Creator Library

Automated Newsletters:

Compose, preview, and dispatch emails directly to your audience mailing lists.

Central Repository:

Search, filter, edit, and cross-post all generated assets from a single searchable library.



Thank You